



Data Visualization in Data Science

Visualization of Information

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Agenda

1. Logistics
2. Goals of the course
3. Introduction – start with the Why?
4. With AI, why do we need visualizations?
5. Some differentiation...
6. Visual Analytics in Machine Learning/Data Science

מאחר שכל השיעורים יועברו בשידור חי בזום ויוקלטו, ציון הקורס יורכב באופן הבא: 60% בחינה ו-40% פרויקט/עבודה. הרכב זה מותנה בחובת נוכחות (בזום או בכיתה) של 80% לפחות.

בהתאם לכך, סטודנטים שלא יוכלו לעמוד במכסת ההשתתפות, בין אם בשל חפיפה עם קורס אחר ובין אם מכל סיבה אחרת, יידרשו להגיש מטלה נוספת שפרטיה ייקבעו בהמשך. ציון המטלה הנוספת יהווה 15% מהציון הסופי.

Please make sure your Zoom name is your full name — attendance is tracked

#	Content	Comment
1	High level overview of the steps of ML and discern when and why visualizations can be used to improve the process	Theory
2	Introduction: Why/What/How visualization pipe, semiology, variables, and taxonomy	Theory
3	Principles of human perception, principles of information processing and perception, saliency, etc.	Theory
4	Data sources and processes of data processing (pre-processing – data transformations)	Theory
5	Visualization methods for temporal data (time-series) and multi-dimensional data (multi variate)	Theory + Hands On
6	Interactive Visualizations with Shiny R (as an example) and other platforms (PowerBI) – how to use gen AI tools to generate visualizations	Theory + Hands On
7	In depth: Principles of interaction, methods of interaction in visualization	Theory + Hands On
8	Lie and truth in information presentation and Dashboards	Theory
9	Visualization of statistical information (e.g. variance, etc.) and visual analytics in machine learning	Theory + Hands On
10	Visualization of text and documents Visualization methods for trees, graphs, and networks	Theory + Hands On
11	Comparison and Evaluation of Visualizations + Story Telling	Theory + Hands On
12	Presentations	
13		

References

Mandatory

- Ward, M. O., Grinstein, G., & Keim, D. (2010). Interactive data visualization: foundations, techniques, and applications. CRC press.
- Foster Provost and Tom Fawcett (2013) - Data Science for Business

Discretionary

- The visual display of quantitative information. Edward Tufte, Graphics press
- Envisioning information. Edward Tufte, Graphics press
- Tamara Munzner. Visualization Analysis and Design. A K Peters Visualization Series, CRC Press, 2014. <https://www.cs.ubc.ca/~tmm/vadbook/>
- Claus O. Wilke, Fundamentals of Data Visualization - A Primer on Making Informative and Compelling Figures, 2019
- Special readings will be provided also for each class/topic.

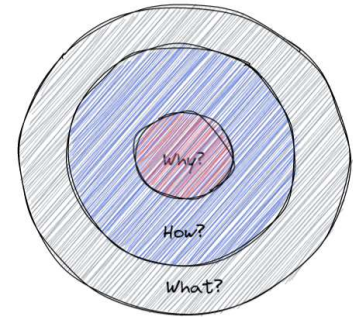
Approach of the course

- Theoretical background on visualization analysis and design.
- Hands-on experience in creating interactive visualizations on real-world data.
- Selected topics on information visualization consisting of time-series, networks, multidimensional attributes and sets visualization.
- Wide range of information visualization techniques.
- Practical approach for the data scientist.

Before we start: Do we still need visualization in the age of AI? [POLL]

- A. Yes
- B. No
- C. Not sure
-  Please vote now in the Zoom poll

Introduction – start with the Why?



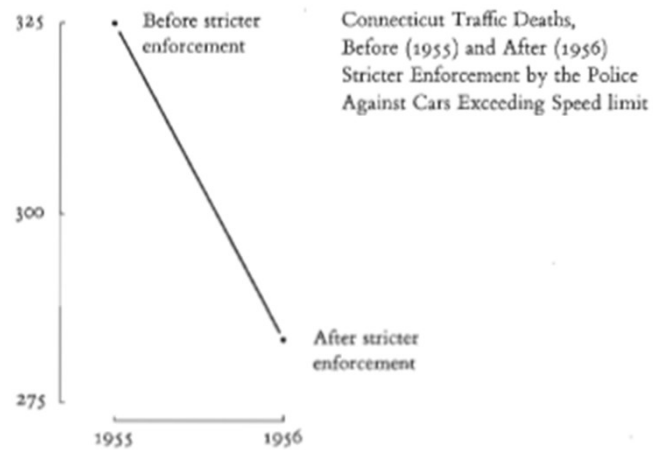
- Image interpretation is performed in parallel within the human perceptual system. (We are visual beings)
- Text analysis is limited by the sequential process of reading.
- Visualizations can be independent of local language, as a graph or a map.
- It is efficient and can be very effective.
- Visualizations are broadly used... so we better learn to use them 😊!



Wikipedia -1913 newspaper advertisement

Is this truthful and revealing?

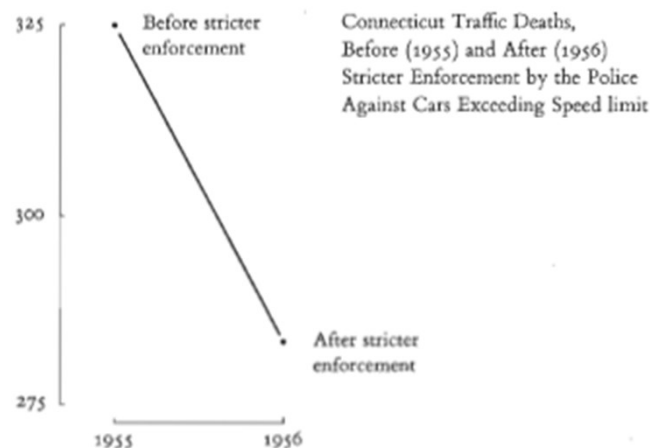
- Let's say the AI gave me this:



Visual Integrity: Context

- To be truthful and revealing, data visualizations should always be able to answer the question: "Compared to what?"

OK now the AI gave me this.



Visual Integrity: Context

- Even better:

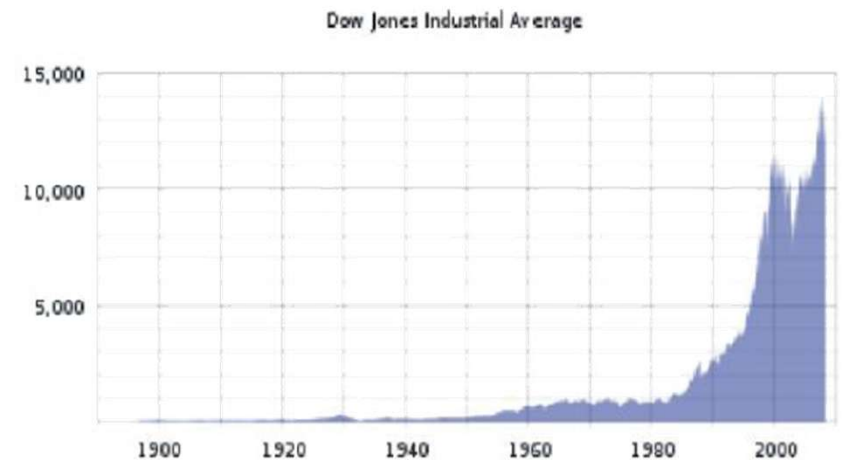


The Importance of Visualizations: The Role of Visualizations in ML and Data Analysis

- Despite advances in AI, human insight (driven by visual interpretation) remains key.
- Visual tools support pattern recognition, anomaly detection, prediction, and discovery.
- Visualizations are essential for understanding data and guiding decisions.
- Our visual system enables fast, intuitive analysis that transcends language.
- Effective visualizations spark insights, questions, and informed decisions are core to EDA.

With AI, why do we need visualizations?

- Be aware there may be humans in the loop!
- Is there a decision-making process?
- Is there an effect of the presentation of data on the decision-making process?
- Can the presentation of data impact the decision?
- Can we say which presentations are better or more influential than others?



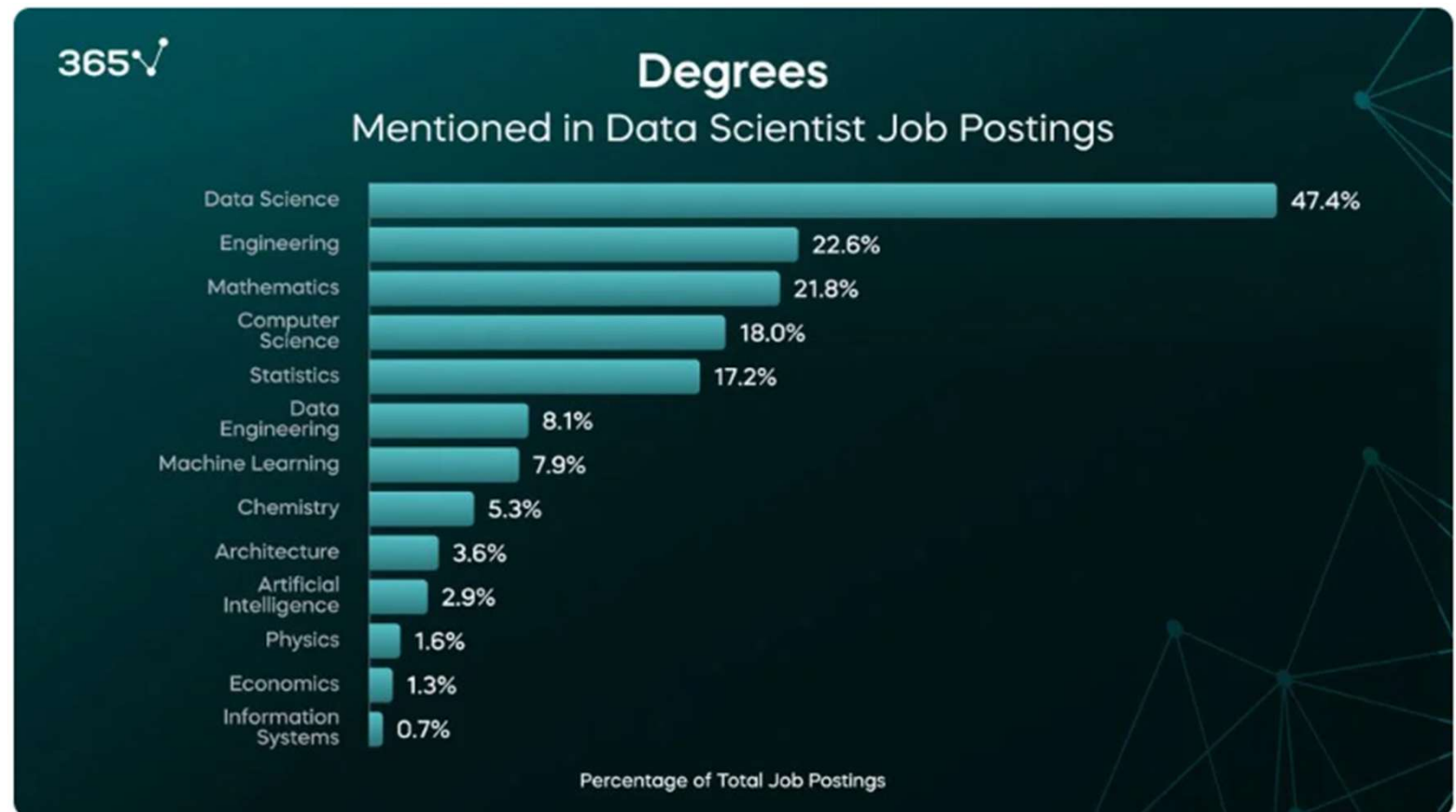
Dow Jones Industrial Average (DJIA) from 1900 to 2000.,
Interactive Data Visualization Second Edition Matthew Ward

With AI, why do we need visualizations?

- Humans want to understand and make sense of the data.
- Humans need to trust!
- The data contain information that, if used effectively, can improve decisions.
- Interaction with visualizations brings new visualizations.

The Importance of Visualizations: The Role of Visualizations in ML and Data Analysis

Despite the rise of ML and AI in data-driven discovery, the ongoing demand for data scientists shows that human insight remains essential in the process.



After this discussion: Do we still need visualization? [POLL]

- A. Yes
- B. No
- C. Not sure
- 👉 Please vote again in the Zoom poll

Some differentiation...

- Scientific Data Visualization vs. Information Visualization
- Visualization vs. Computer Graphics

Several Data Sets with the same statistics:

Viewing data

Anscombe, F. J. (1973).
Graphs in Statistical
Analysis. *American
Statistician* 27 (1): 17–21.

Mean X	9
Variance X	11
Mean Y	7.5
Variance Y	4.12
Correlation x, y	.816
Linear function	$y = 3 + 0.5x$

Will these datasets look similar when plotted?
[POLL]

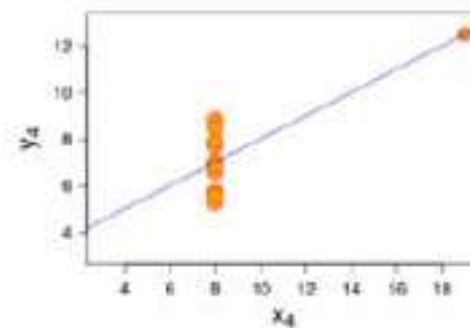
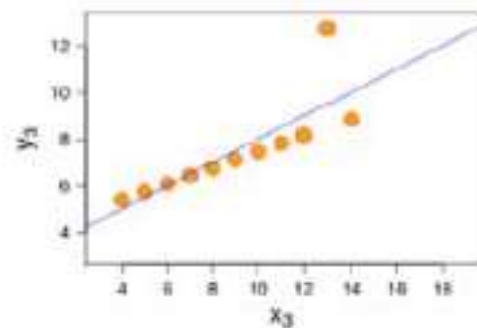
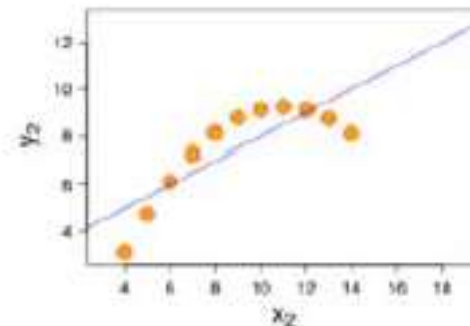
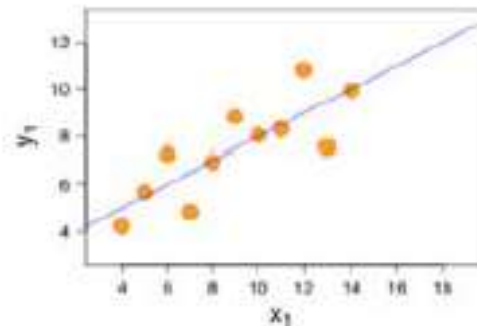
- A. Yes
- B. No
- 👉 Please vote now in the Zoom poll

Statistics is not the data

Viewing data

Anscombe, F. J. (1973).
Graphs in Statistical
Analysis. *American
Statistician* 27 (1): 17–21.

Mean X	9
Variance X	11
Mean Y	7.5
Variance Y	4.12
Correlation x, y	.816
Linear function	$y = 3 + 0.5x$



Anscombe quartet data

x123	y1	y2	y3	x4	y4
10	8.04	9.14	7.46	8	6.58
8	6.95	8.14	6.77	8	5.76
13	7.58	8.74	12.74	8	7.71
9	8.81	8.77	7.11	8	8.84
11	8.33	9.26	7.81	8	8.47
14	9.96	8.1	8.84	8	7.04
6	7.24	6.13	6.08	8	5.25
4	4.26	3.1	5.39	19	12.5
12	10.84	9.13	8.15	8	5.56
7	4.82	7.26	6.42	8	7.91
5	5.68	4.74	5.73	8	6.89

What?

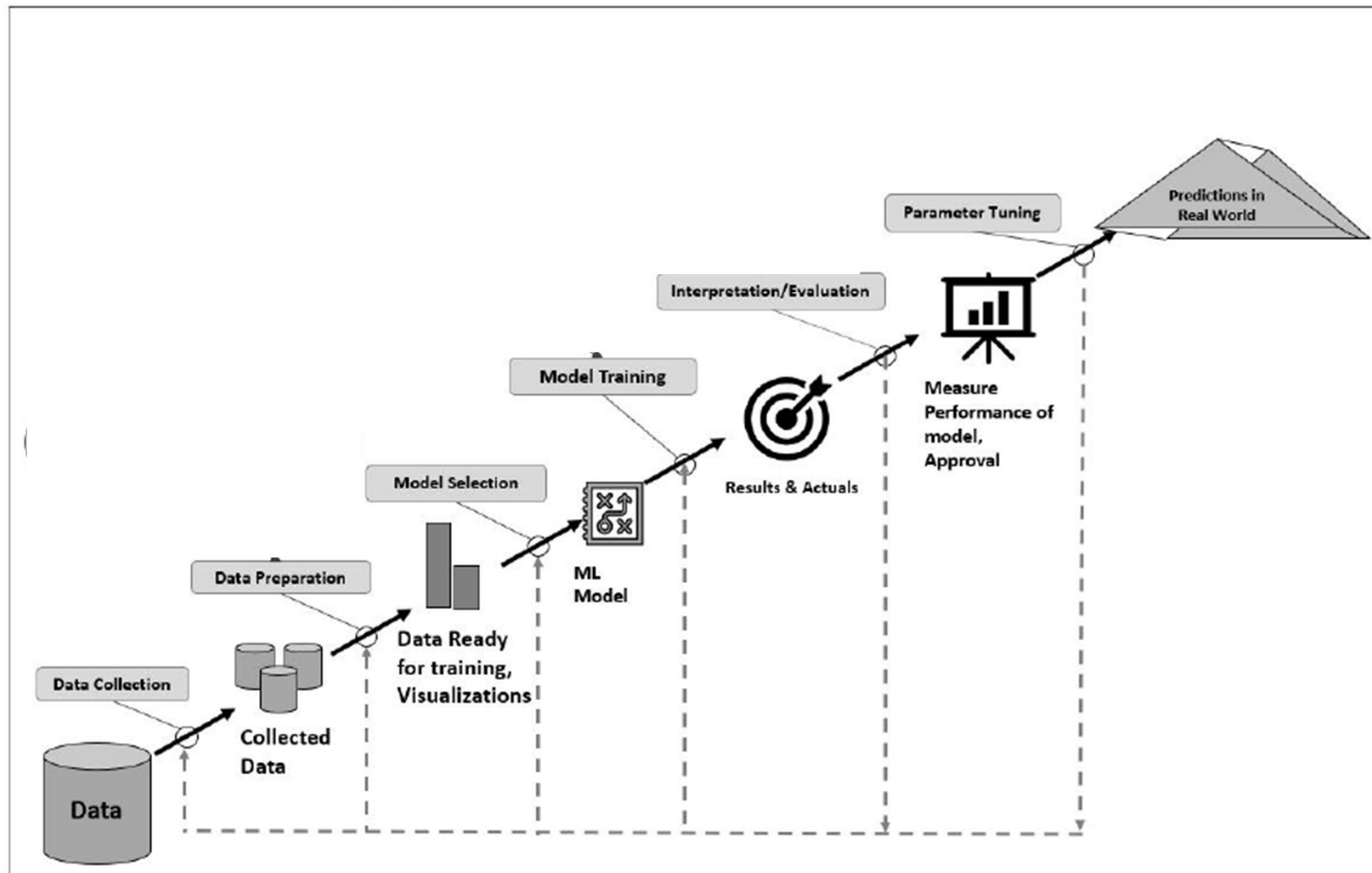
- We define *visualization* as the **communication** of information using graphical representations of data.
- Or: Visualization is a process that **transforms information into a visual representation**, enabling the viewer to observe, browse, make sense and understand information. (www.infovis.org)
- We need to understand here:
 - Communication
 - Information
 - Representation
 - Data

Is this a visualization?

x123	y1	y2	y3	x4	y4
10	8.04	9.14	7.46	8	6.58
8	6.95	8.14	6.77	8	5.76
13	7.58	8.74	12.74	8	7.71
9	8.81	8.77	7.11	8	8.84
11	8.33	9.26	7.81	8	8.47
14	9.96	8.1	8.84	8	7.04
6	7.24	6.13	6.08	8	5.25
4	4.26	3.1	5.39	19	12.5
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Anscombe quartet

Visual Analytics in Machine Learning



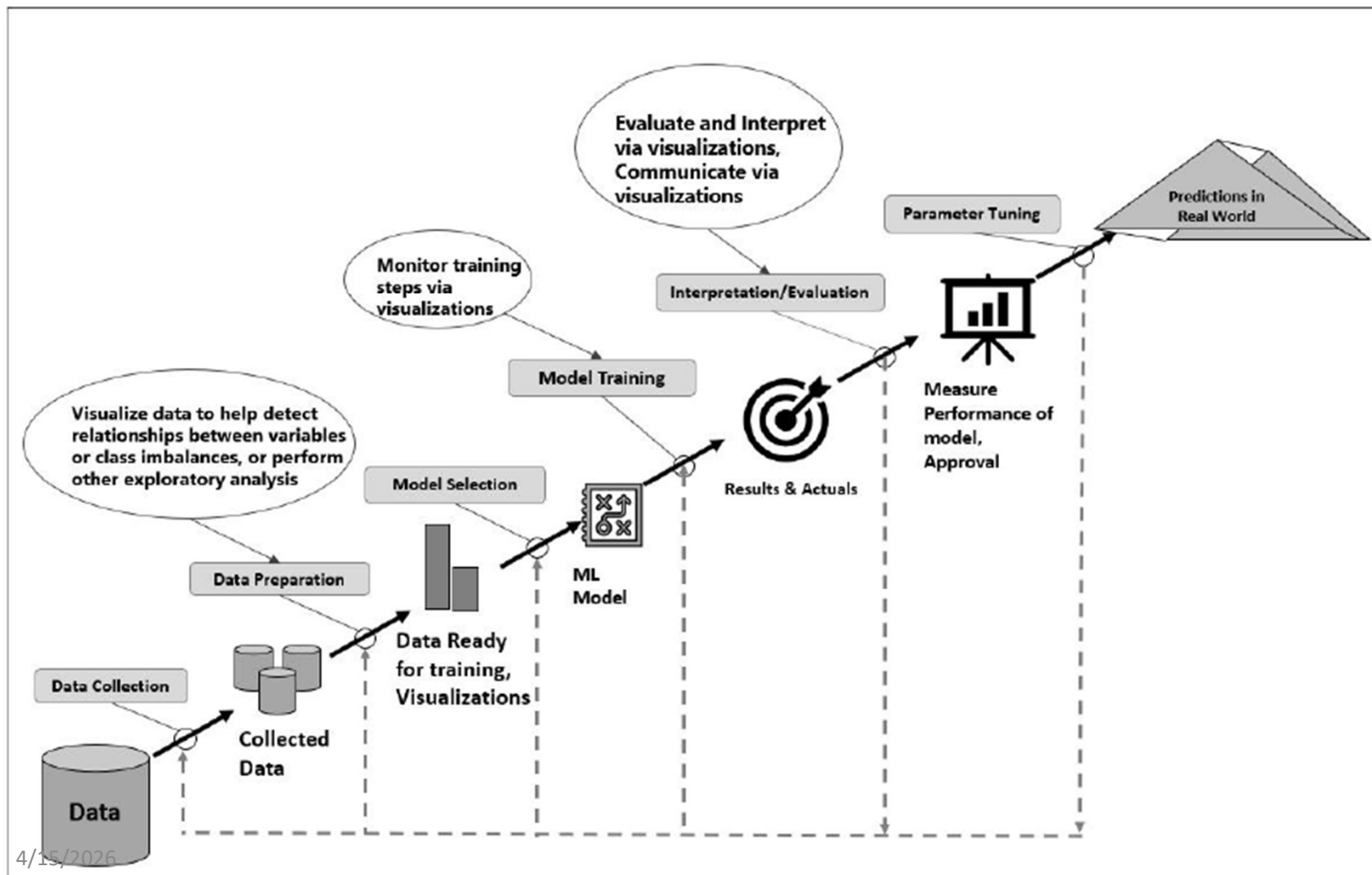
Yufeng Guo.:
The 7 Steps of
Machine
Learning, 2017

Where is visualization MOST important?

[POLL]

- A. Data preparation
 - B. Model training
 - C. Evaluation
 - D. Everywhere
-
- 👉 Please vote now in the Zoom poll

Visual Analytics in Machine Learning



Yufeng Guo.:
The 7 Steps of
Machine
Learning, 2017

The 7 Steps of Machine Learning

1. Data Preparation: process of converting the data into the required structure

- Transforming the data to a tabular format, removing or inferring missing values, removing outliers and anomalies and converting data to different types are examples of data preparation.
- Often involve data visualizations to help detect relevant relations between variables or class imbalances, identify anomalies and outliers and perform other exploratory analyses.

The 7 Steps of Machine Learning

1. Data Preparation: process of converting the data into the required structure

- Visualizations should not be very different from the visualizations used in classical KDD (knowledge data discovery).
- Exception: often involve large amounts of data, and the data scientist can encounter visual scalability issues.

The 7 Steps of Machine Learning

2. Model Selection: it is determined by the business or scientific question that has to be answered using ML, the type of data and its behavior.

- Often more than one model can answer the question.
- **Need to understand the data to select the model**, and visualizations may support this task.
- Even if visualizations are not used in this step, it is important to review this step, because the selected model may also determine the visualizations used in the next steps of the process.

The 7 Steps of Machine Learning

3. Model Training: the model learns by using the data and computes its internal parameters.

- Done on a set of data that should be as **ergodic** as possible
- Visualizations are important, as they can help to monitor how the model converges to a solution.

The 7 Steps of Machine Learning

4. Evaluation: assess the ML results rigorously to gain confidence that they are valid and reliable and that the model satisfies the original business goals before moving on.

- Visualizations can support this step very efficiently.

The 7 Steps of Machine Learning

5. Interpretation: involves qualitative assessments.

- Various stakeholders have interests in the decision-making that will be accomplished or supported by the resultant models.
- It is important to consider the comprehensibility of the model to stakeholders
- Visual analytics may be crucial.

The 7 Steps of Machine Learning

6. Parameter Tuning: is more an art than a technique.

- Objective is to improve performance through fine tuning of the number of training steps, learning rate, initialization values and distribution, etc.
- Hyperparameters values configure characteristics of the model and may highly impact the training performance

The 7 Steps of Machine Learning

6. Parameter Tuning: is more an art than a technique.

- given the complexity of the model algorithms and the training processes, identifying a sweet spot in the hyperparameter space for a specific problem can be challenging
- Dashboards of SPLOM, heatmaps and line plots can be used to determine the optimal set of parameters. (SPLOM = Scatter Plot Matrix)

The 7 Steps of Machine Learning

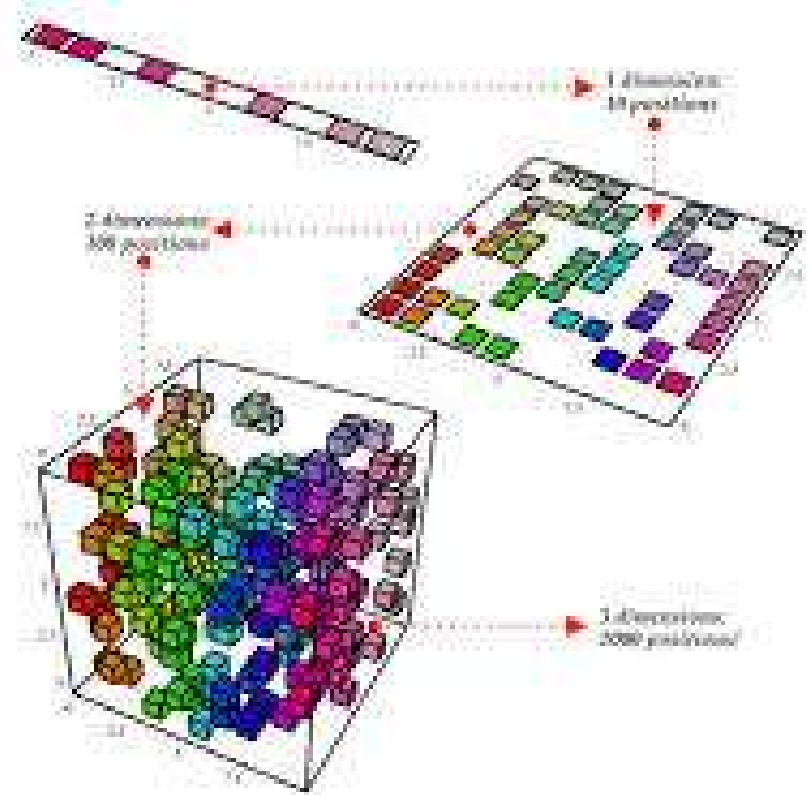
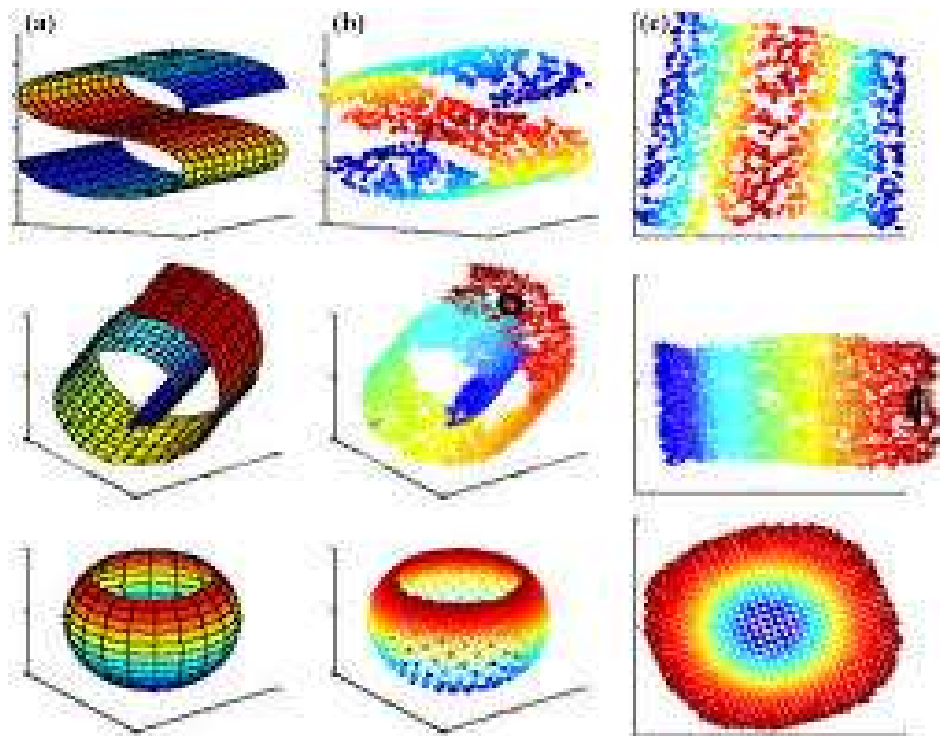
7. Prediction Making: monitoring visualizations
Dashboards

1. Data Preparation

Scalability Problem:

- Very large number of observations and number of features
- Harder to provide clear and easy to understand visualizations.
- Automatic analysis methods to extract potentially relevant visual structures
- Data perspective: the most popular automatized available tool for simplifying visualizations and for providing a better understanding is Dimensionality Reduction

Dimensionality Reduction



Data: Networks (graphs) and Trees

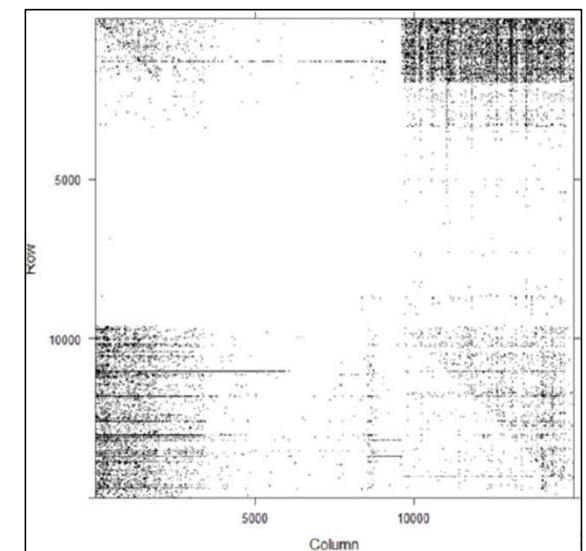
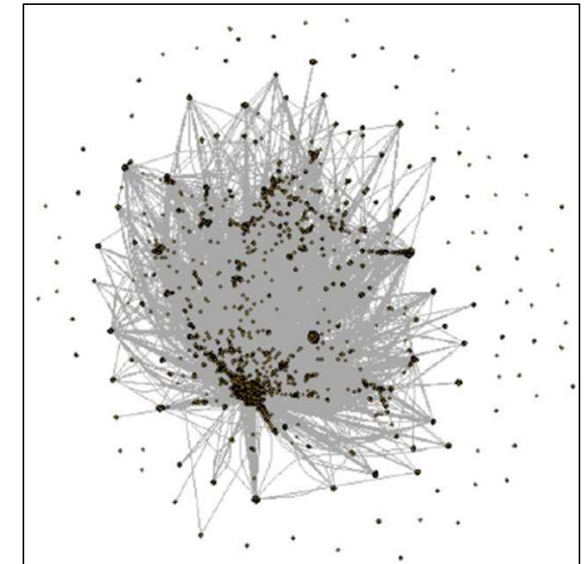
- Node is an item in a network
- Link or edge is a relation between two items.
- Examples:
 - Social Network: the nodes are people, and links mean friendship.
 - Gene interaction network: the nodes are genes, and links between them mean that these genes have been observed to interact with each other.
 - Computer network: the nodes are computers, and the links represent the ability to send messages directly between two computers using physical cables or a wireless connection.
- Nodes can have associated attributes, just like items in a table.
- Links can have associated attributes and may be partly or wholly disjoint from the node attributes.
- A graph (network) has loops/cycles inside
- A tree is graph which does not have loops/cycles inside

Data Preparation

Example: Network data that is visualized through graphs

- First step is to convert the graph data into a sparse matrix (called an adjacency matrix)
- Second step is usually a matrix reorganization to reduce the graph to smaller graphs by partitioning its nodes into mutually groups.

Graph visualization of 15000 nodes from an Email network, based on traffic data collected for 112 days. Graph visualization; right, adjacency matrix using the R igraphs package



Network data - examples

email.edgelist1

V1	V2
1	63
1	144
1	196
1	384
1	10589
1	14267
1	14706
6	125
6	9826
6	10970
6	12604
6	12838
6	13111
12	327
12	11982
12	14975
13	11798
13	12300
13	12338
14	14277
15	970

Data: Networks (graphs) and Trees

```
> head(original_edgelist)
```

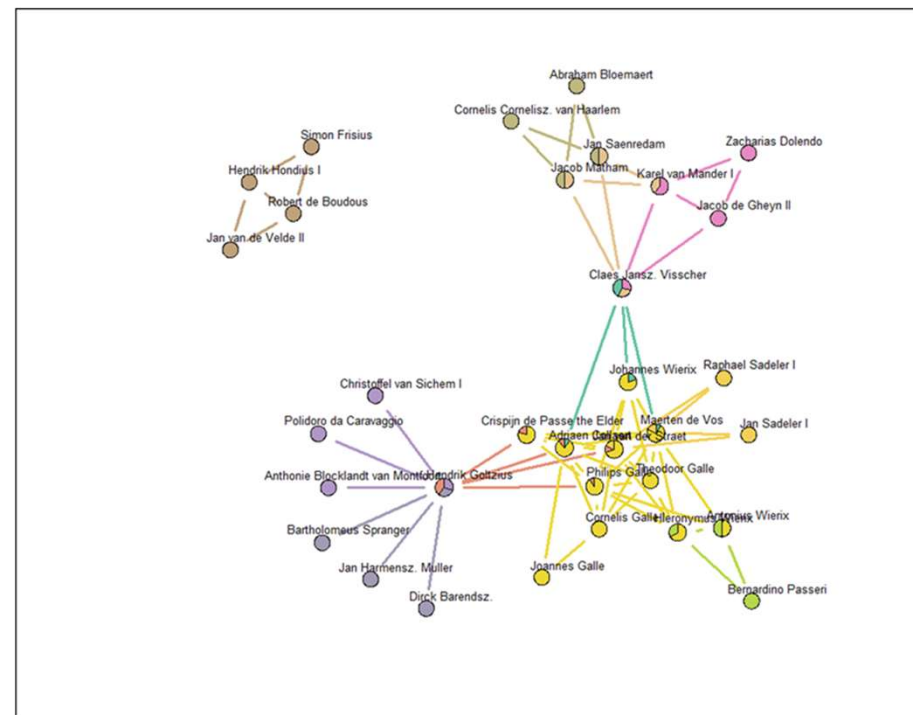
	from	to	weight
1	Jan van der Straet	Philips Galle	494
2	Maerten de Vos	Philips Galle	80
3	Johannes Wierix	Philips Galle	37
4	Theodoor Galle	Philips Galle	91
5	Hendrik Goltzius	Philips Galle	43
6	Crispijn de Passe the Elder	Philips Galle	43

```
> head(original_nodelist)
```

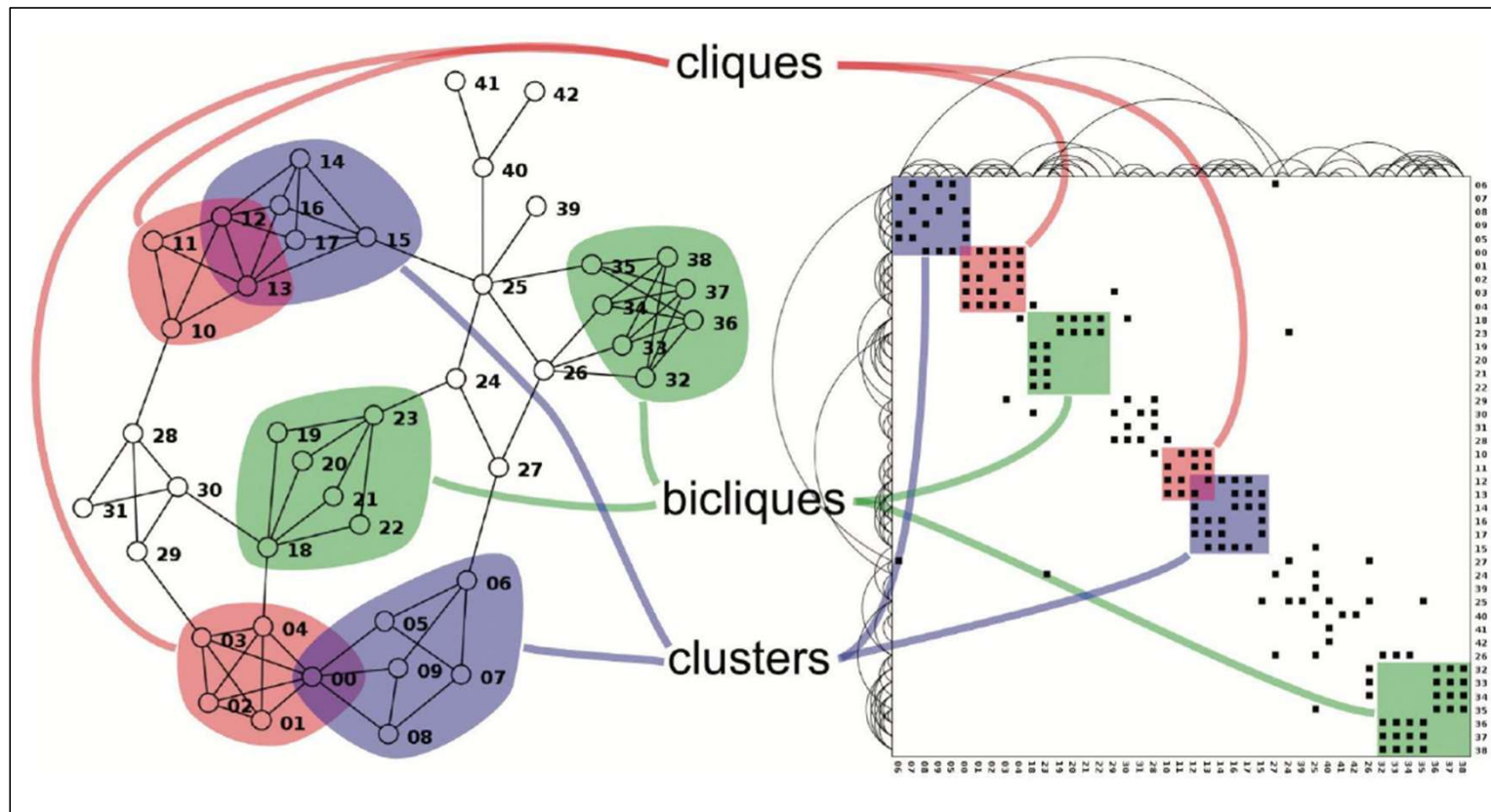
	name	birth	nationality
1	Philips Galle	1537	Flemish
2	Jan van der Straet	1523	Flemish
3	Raphael Sadeler I	1560	Flemish
4	Maerten de Vos	1532	Flemish
5	Albrecht Dürer	1471	German
6	Johannes Wierix	1539	Flemish

```
> head(dd2)
```

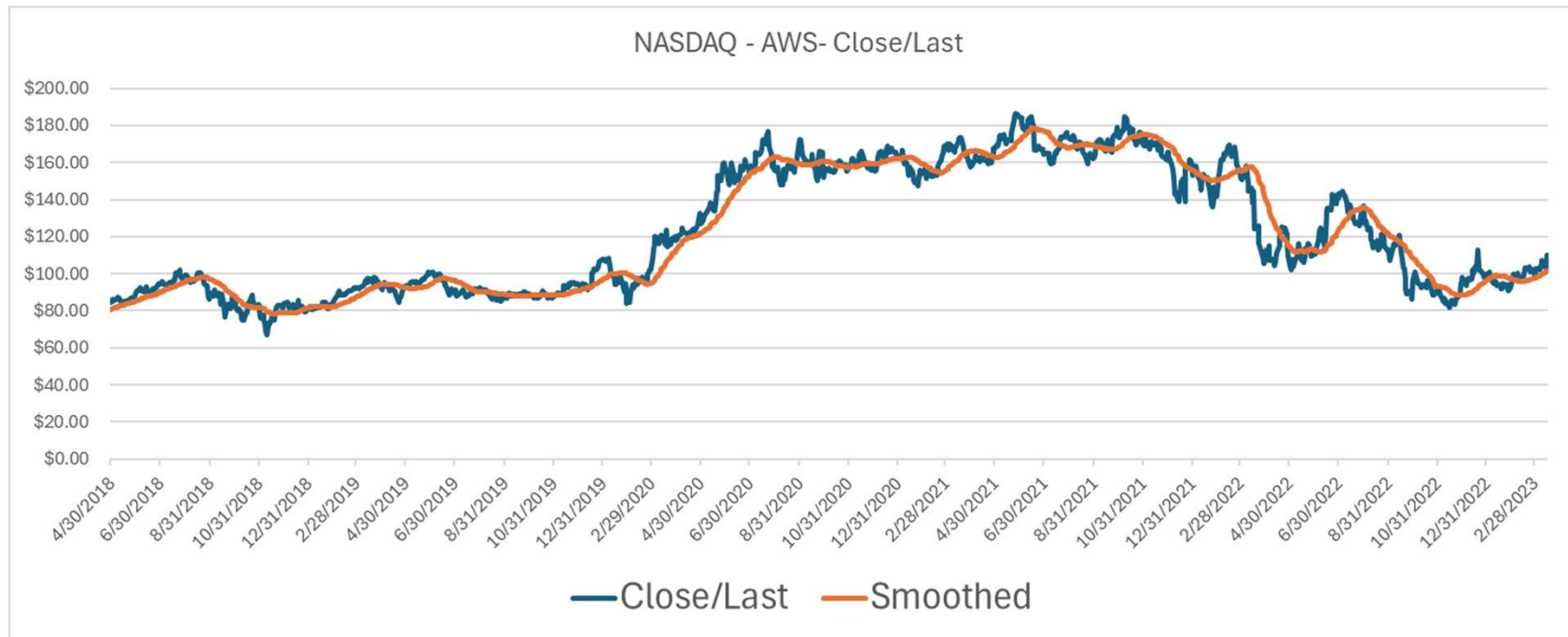
	v1	v2
1	15	9721
2	15	10201
3	23	14548
4	23	12115
5	23	10161
6	23	9935



Graphs vs. Matrix



Time Series



Types of data preparation/transformation:

- Seasonality reduction
- Smoothing by moving averages

Choosing a model

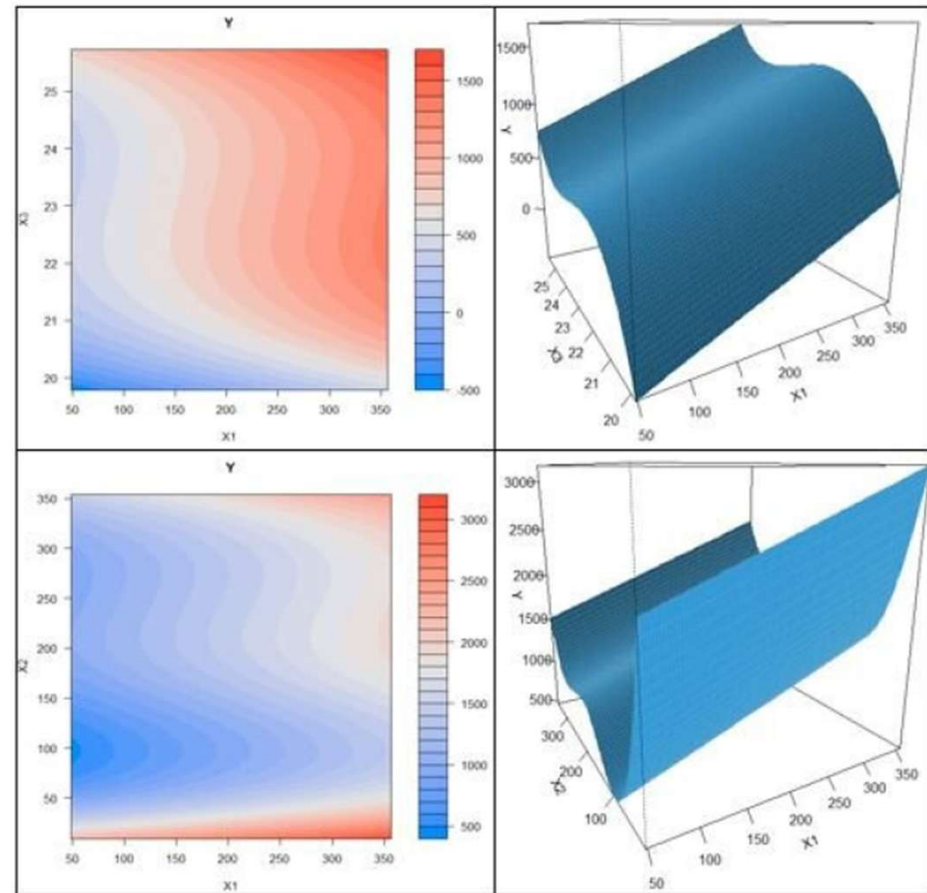
- Numerous supervised and unsupervised ML methods exist and require different visualization techniques.
- Visualization techniques for a specific Machine Learning model depend on the the data-types and the selected model.
- Data scientists mainly use simple and intuitive visualizations: line plots, scatter plots, heat-maps, contour plots, hierarchical trees and several combinations of them.

Choosing a model - supervised

Representations of the **regression** surface as a function of X_1 , X_2 and X_3 , using synthesized data where $Y = 22 + 3X_1 + 2X_2 + 3X_3$.

Left: Filled contour plots.

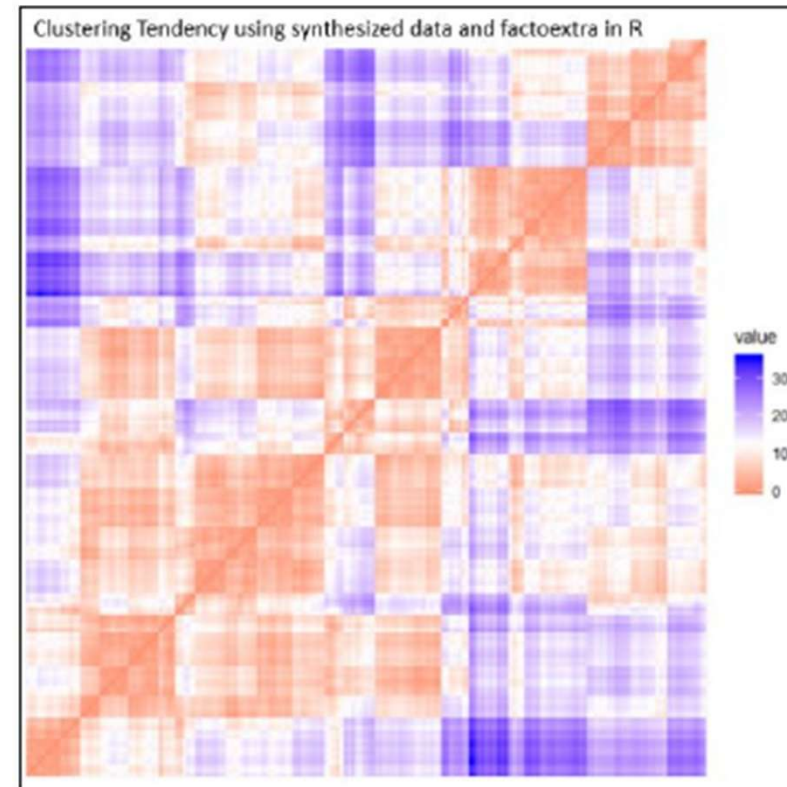
Right: Perspective plots.



Choosing a model - unsupervised

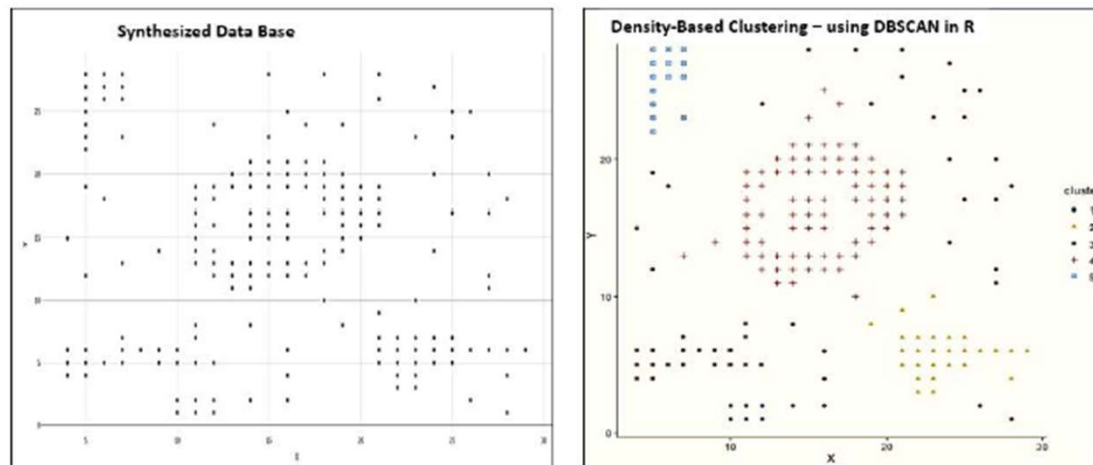
- Before applying clustering algorithms to a data set, first assess the clustering tendency (to understand whether the data set has a natural clusters or not)
- Classification of objects into clusters requires methods for measuring the distance or the (dis)similarity between the objects.
- Visualizations can help to expose and analyze both the clustering tendency and similarity distances in the data
- Clustering tendency is detected in a visual form by counting the number of square shaped dark blocks along the diagonal in the image.
- Red: high similarity (ie: low dissimilarity) , Blue: low similarity.

4/15/2026



Choosing a model - unsupervised

- Density based clusters are an additional way to visualize possible clusters.
- Main reason why it is possible to recognize clusters here is that the density of points within each cluster is clearly higher than the density outside the cluster

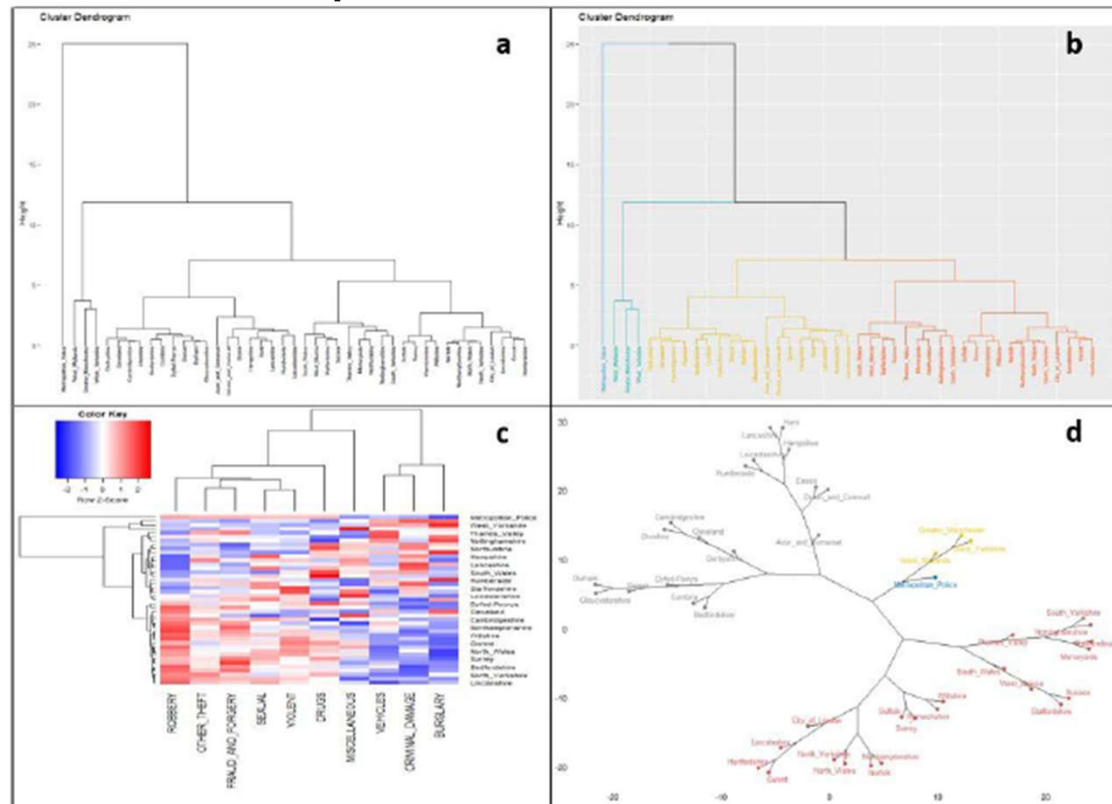


Density based clustering can help to find clusters with different shapes and sizes from data containing noise and outliers

Choosing a model - unsupervised

- Visual representation of hierarchical clustering is a tree-based visualization of the objects, which is also known as dendrogram
- It can also be used to analyze network data.

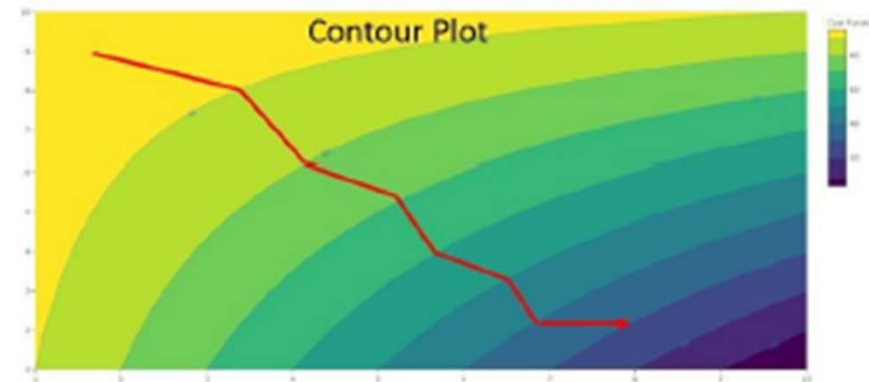
Offences recorded by the police in England and Wales by offence and police force area for 2001/02, from <https://www.gov.uk/government/statistics/historical-crime-data>



a) Simple dendrogram, b) dendrogram with 4 groups, c) Heat Map using package pheatmap d) Phylogenetic dendrogram.

Training the model

- Goal of training is to enable the machine to learn the data, so that the answers to questions or the prediction are as correct as possible.
- Needs to monitor the algorithm iterations to verify convergence and to evaluate the results.
- The progress of gradient descent on a test surface can be visualized for all the steps.
- Can be useful to display the three-dimensional data in two dimensions, using contours or color-coded regions



The red line shows the convergence of the process to the minimum of the cost function.

Evaluation

- Human interaction is very important in the evaluation step of the process.
- The right data, combined with the right data science techniques, will help to identify the models that optimize a cost criterion.
- However, only humans can decide on what is the best criterion
- One strategy for decision making is to rank a set of cases by scores, and then select the cases at the top of the list.

Evaluation -Profit Curves

Definition: A **profit curve** measures the expected profit (or utility) of using a classifier at different decision thresholds. It considers:

- **True positives (TP), False positives (FP), True negatives (TN) and False negatives (FN).**
- **Cost/benefit structure:** The economic or operational value associated with each type of prediction.

Key Idea:

- Profit curves help in selecting the optimal threshold for classification by **maximizing profit** rather than just improving accuracy.
- They are particularly useful when the conditions under which the classifier will be deployed are well understood (e.g., fraud detection in banking, targeted marketing).

Evaluation -Profit Curves

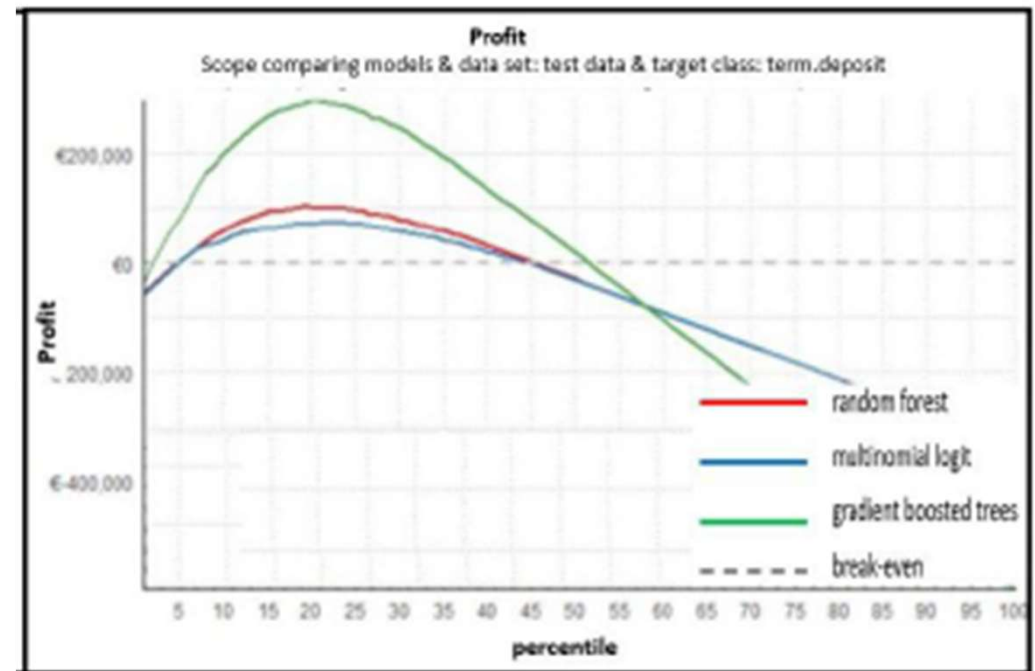
understood (e.g., fraud detection in banking, targeted marketing).

How to Interpret:

1. The **x-axis** represents the proportion of selected cases (typically ranked by predicted probability).
2. The **y-axis** represents the profit associated with that selection.
3. The peak of the profit curve indicates the **optimal cutoff**—the point where profit is maximized.

Example Use Case:

- In a **marketing campaign**, sending promotional emails to potential customers involves costs. A profit curve helps determine the optimal number of emails to send before the cost outweighs the benefit.



profit curves for three classifiers, using R package modelplotr based on a Bank Marketing Data Set

Evaluation - Cumulative Gains

- **Definition:** Cumulative gains measure how well a classifier ranks positive instances compared to a random selection. This is often visualized using a **Gains Chart**.

Key Idea:

- It helps answer: "**If we target the top X% of ranked cases, what percentage of actual positives do we capture?**"
- More effective classifiers will **capture a higher percentage of positives** earlier in the ranking.

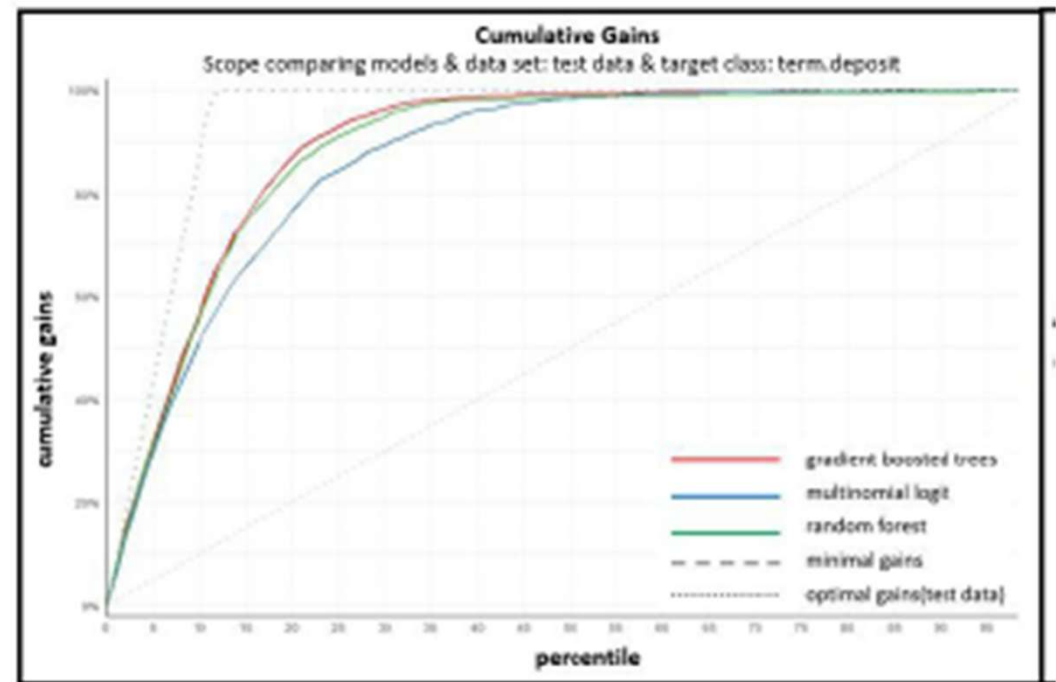
Evaluation - Cumulative Gains

How to Interpret:

1. The **x-axis** represents the percentage of the ranked population considered.
2. The **y-axis** represents the percentage of true positives captured.
3. A **random classifier** (baseline) would have a diagonal line (capturing positives at the same rate as selecting randomly).
4. A **better classifier** will have a steep curve, capturing a high proportion of positives early.

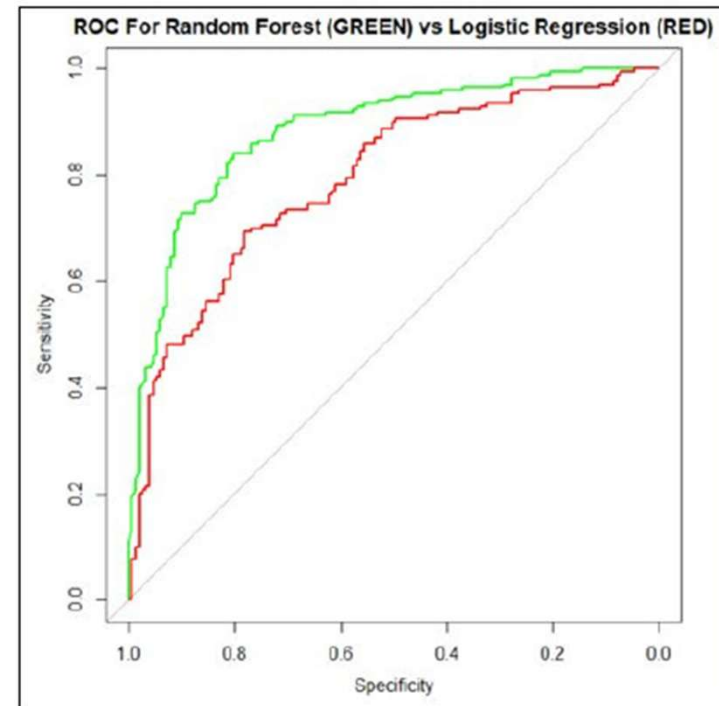
Example Use Case:

- In **fraud detection**, a model that correctly ranks high-risk transactions at the top will capture most fraudulent cases with fewer investigations.



Evaluation - ROC

- Different evaluation methods should be used when the conditions under which the classifiers are used are uncertain or unstable.
- Example: Receiver Operating Characteristics (ROC) curve.
- ROC is a probability curve and Area Under the Curve (AUC) represents the degree of how much the model is capable of distinguishing between classes.
- The higher the AUC, the better the model predicts 0s as 0s and 1s as 1s or distinguishes between positives and not positives



Interpretation and explainability – why it is important.

- Highly debated topics in deep learning is how to interpret and understand a trained model, particularly in the context of high-risk industries, such as healthcare.
- Visualizations in this step are critical, as they can be crucial in getting the organization's trust to accept and adopt the specific ML model and algorithms.
- Visualization methods provide the necessary means to simultaneously analyze the huge amount of information hidden in a deep learning neural network.

Interpretation

- Typify ML models interpretability into two types according to the model's interpretability: **glass-box** and **black-box** models.
 - Interpretable by design, such as linear models, rule lists, decision trees, regressions, generalized additive models.
 - Interpretability was not an intrinsic element of their design (such as CNNs) and that need an additional “ML” layer that uses explainability methods (i.e.: Partial Dependence, LIME, SHAP) for explaining those models

Interpretation

Two very known methods are Shapley Values and LIME.

- LIME tests what happens to the predictions when there are variations in the features data of the ML model, which is a very intuitive approach. (**Local Interpretable Model-agnostic Explanations**)
- Shapley value, might be the only method to deliver a full explanation for ML models, while methods like LIME only assume linear behavior of the ML locally which has not been theoretically proven

Comparison of SHAP and LIME Methods

Key Differences:

1.Theoretical Basis:

1. SHAP is rooted in game theory and provides a rigorous framework for feature attribution.
2. LIME is heuristic-based and focuses on approximating the model locally.

2.Explanation Scope:

1. SHAP can explain both individual predictions (local) and the overall model behavior (global).
2. LIME is primarily designed for local explanations.

3.Consistency:

1. SHAP guarantees consistency (if a feature's contribution increases, its Shapley value won't decrease).
2. LIME does not provide such guarantees and may produce inconsistent explanations.

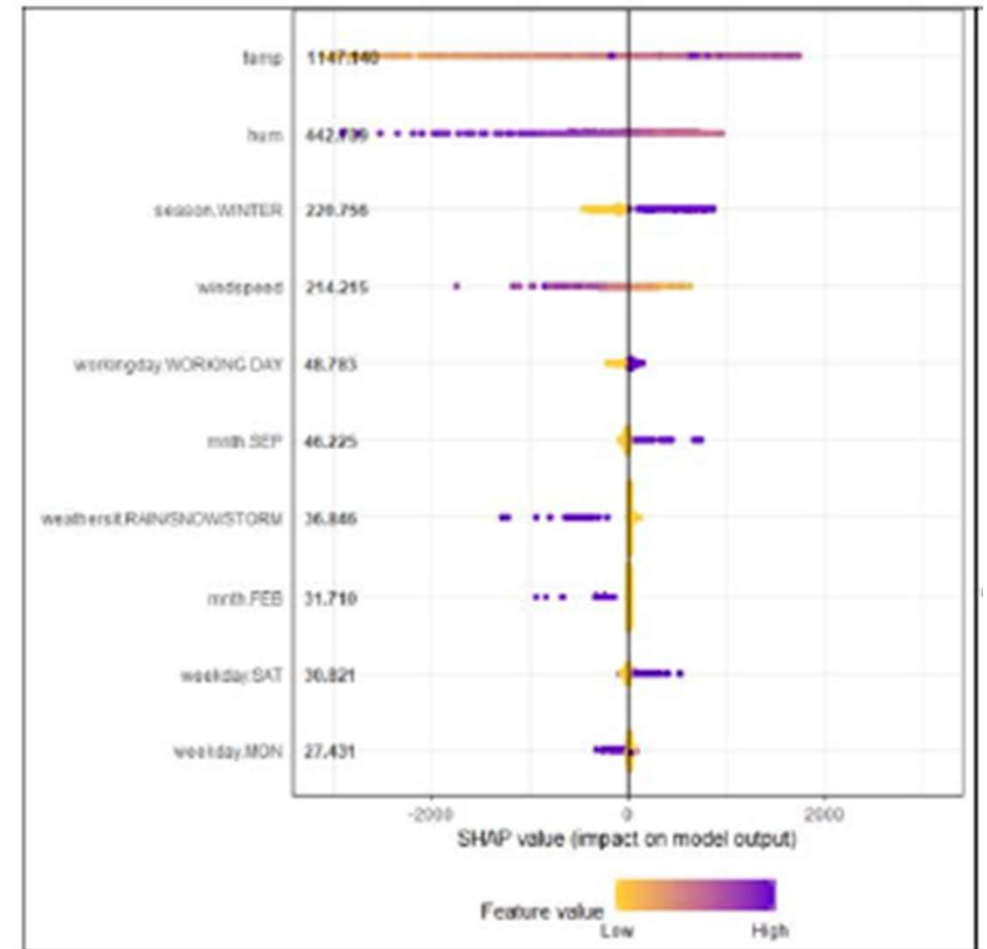
4.Performance:

1. SHAP can be slower due to its computational complexity.
2. LIME is generally faster but may sacrifice some accuracy in explanations.

SHAP summary plots

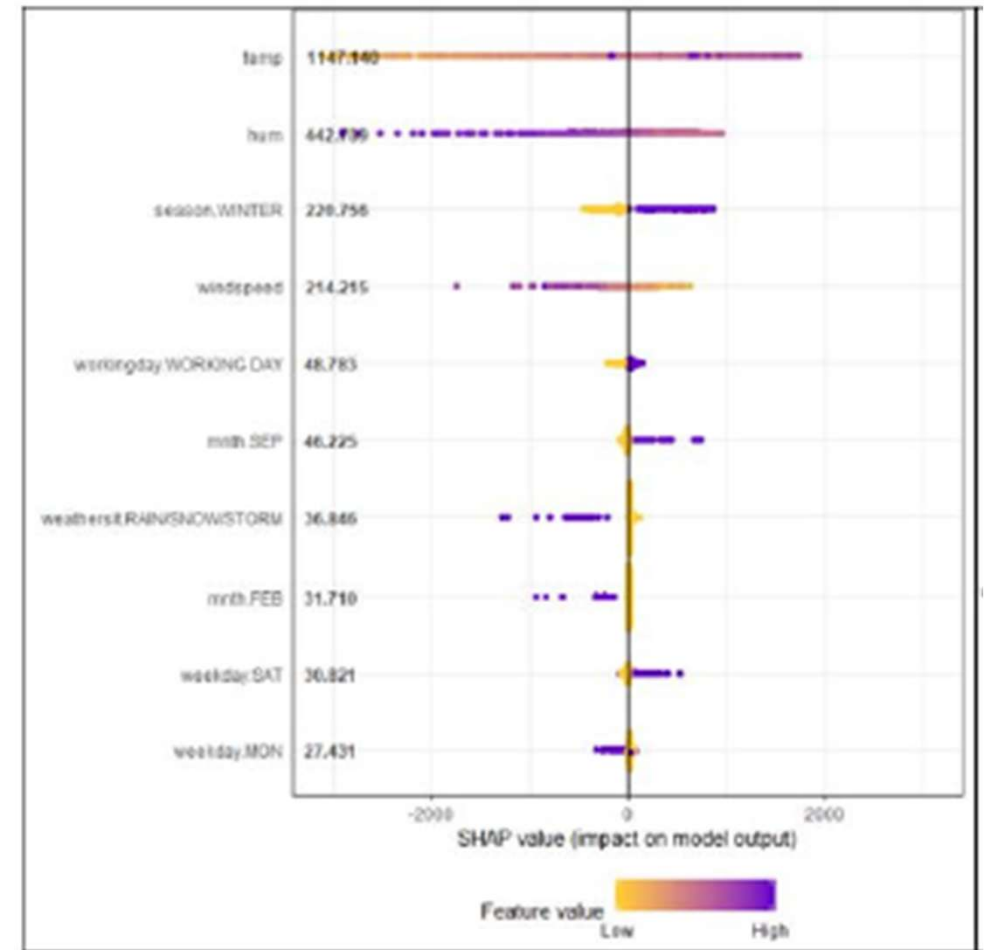
SHAP: SHapley Additive exPlanations

- SHAP summary plot of a 10-feature model. The y-axis indicates the variable names, in order of importance from top to bottom.
- The value next to them is the mean SHAP value.
- On the x-axis is the Shapley value, which indicates the change in log-odds.
- From this number, it is possible to extract the probability of success.
- Gradient color indicates the original value for that variable.
- Each point represents a row from the original data-set



SHAP summary plots

1. Position (ranking) → ground truth
2. Density away from zero → consistency
3. Spread → variability (not importance alone)



Taxonomy - visual analytics and the 7 steps of ML

	Model	Data preparation - Exploration	Training	Evaluating	Interpretation	Hyperparameters tuning
Classic Supervised	Classification trees	Hierarchical Plots	Tree Network: Hierarchical Plots	Confusion Matrix: Heat Maps ROC plots: line plots Cumulative response curves: line plots Profit curves: line plots	Predictors Importance: Classical bar chart ; SHAP summary plot Predictors inter-dependence: classical dependence line plots, Classical 3D dependence plot, SHAP dependence plots SHAP Interaction effects plots	
	Decision Tree Inducers		Confusion Matrix Evolution Plot			
	Bayesian networks	Dimension Reduction ,Clustering Tendency: SPLOM Cluster Tendency: Map Heat Density Based Cluster: Scattered Plot Similarity-Dissimilarity: Scattered Plot	Gradient Descent: Contour Plot (3D and 2D) Confusion Matrix Evolution Plot			
	Regressions		Decision Boundary: Scattered Plot Confusion Matrix Evolution Plot			
Classic Unsupervised	Support Vector Machines (SVM)					Capture the Hyperparameters and KPIs for each tuning round (KPIs are different for the different models) Combine SPLOM, Line plots and Heatmaps. Slice and dice the data to identify the best combination of parameters for the relevant KPIs.
	Cluster		Cluster Tendency: Map Heat Density Based Cluster: Scattered Plot Similarity-Dissimilarity: Scattered Plot	Silhouette Coefficient: Clusters silhouette plot Similarity-Dissimilarity: Scattered Plot	Scatter Plots	
	Network Data - Graphs/Link Analysis	Graph plots with color density Motifs: aggregation Spectral Partitioning / Scale XY layout	Graph plots with color density Scatter plot matrices	Graphlet frequencies for similarity	Motifs using glyphs Motifs modules Graphlet frequencies for similarity	
Deep Learning	Neural Networks; Genetic; CNN, Recurrent neural network, Deep belief networks	Dimension Reduction ,Clustering Tendency: SPLOM Cluster Tendency: Map Heat Density Based Cluster: Scattered Plot Similarity-Dissimilarity: Scattered Plot	Gradient Descent: Contour Plot (3D and 2D) Cluster Tendency: Map Heat Density Based Cluster: Scattered Plot Similarity-Dissimilarity: Scattered Plot Confusion Matrix Evolution Plots Validation loss per step: line plots Validation accuracy per step: line plots Confusion Matrix Evolution Plots	Confusion Matrix: Heat Maps ROC plots: line plots Cumulative response curves: line plots Profit curves: line plots	Scatter Plots Predictors Importance: Classical bar chart , SHAP summary plot Predictors inter-dependence: classical dependence plots, SHAP dependence plots SHAP Interaction effects CARTscan scatter plots Display activation: Image Matrix or Heat Map	
Ensemble Methods	Boosted trees, Bagged trees					

The assignment /Project

- In groups of 3..
- Not theoretical – try to choose a model that you are developing or will develop
- Needs data
- Needs coding
- If AI is used, need to show the method and prompts used to develop
- Related to the 7 Steps of Machine Learning